GIT Basics

- GIT Basics covers the fundamental commands required to start working with a project using GIT. These commands help a developer create a repository, check file status, prepare changes, save changes as commits, and view the history of committed changes.

1. GIT Installation

Command

```
git --version
```

Purpose of the Command

- Check whether GIT is installed and displays the installed GIT version.

Example

```
git --version
```

Example output:
git version 2.x.x

2. GIT Configuration

Command

```
git config --global user.name "Your Name"
git config --global user.email "your@email.com"
```

Purpose of the Command

- Configures the name and email associated with commits.

Example

```
git config --global user.name "Rahul Kumar"
git config --global user.email "rahul@example.com"
```

3. GIT Initialization

Command

```
git init
```

Purpose of the Command

- Creates a new local GIT repository in the current project folder.

Example

```
cd StudentManagement
git init
```

4. GIT Status

Command

```
git status
```

Purpose of the Command

- Shows the current state of the working directory and staging area.

Example

```
git status
```

5. GIT Staging

Staging selects changes that should be included in the next commit. The staging area is between the working directory and the repository.

A. Adding a Single File

Command

```
git add Student.java
```

Purpose

Stages only Student.java.

B. Adding Multiple Files

Command

```
git add Student.java Course.java
```

Purpose

Stages the specified files.

C. Adding All Files

Command

```
git add .
```

Purpose

Stages all applicable changes in the current directory and its subdirectories.

6. GIT Commit

Command

```
git commit -m "Commit message"
```

Purpose of the Command

- Records the currently staged changes in the local repository.

Example

```
git commit -m "Added student class"
```

7. GIT Commit History

Detailed History

Command

```
git log
```

Purpose of the Command

- Displays the commit history of the current repository.

Example

```
git log
```

Short History

Command

```
git log --oneline
```

Purpose

Displays each commit in a compact single-line format.

GIT Basics – Command Summary

Command	Purpose
git --version	Checks installed GIT version.
git config	Configures GIT settings.
git init	Creates a local GIT repository.
git status	Shows repository/worktree state.
git add <file>	Stages one file.
git add file1 file2	Stages selected files.
git add .	Stages all applicable changes.
git commit -m	Records staged changes.
git log	Shows detailed history.
git log --oneline	Shows compact history.